

Brutal Series - Paper B13: Respiration / Lines & Angles / Soil / Algebraic Expressions

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A person breathes 18 times a minute at rest, but 30 times a minute during exercise. How many MORE breaths are taken in 5 minutes of exercise than in 5 minutes of rest? [\[Respiration in Organisms\]](#)
 - (A) 60 more breaths
 - (B) 12 more breaths
 - (C) 48 more breaths
 - (D) 150 more breaths
 - (E) 240 more breaths
 - (F) 75 more breaths

2. Two angles are complementary and one of them is 37 degrees. The other angle then forms a linear pair with a third angle. What is that third angle? [\[Lines & Angles\]](#)
 - (A) 117 degrees
 - (B) 90 degrees
 - (C) 143 degrees
 - (D) 233 degrees
 - (E) 37 degrees
 - (F) 53 degrees
 - (G) 127 degrees

3. In a percolation test, 200 mL of water drains through a soil sample in 20 minutes. What is the percolation rate in mL per minute? [\[Soil\]](#)
 - (A) 0.1 mL per minute
 - (B) 10 mL per minute
 - (C) 180 mL per minute
 - (D) 20 mL per minute
 - (E) 40 mL per minute
 - (F) 100 mL per minute
 - (G) 4000 mL per minute

4. Find the value of $3x^2 - 2x + 5$ when $x = 4$. [\[Algebraic Expressions\]](#)
 - (A) 48
 - (B) 45
 - (C) 33
 - (D) 53
 - (E) 40
 - (F) 9
 - (G) 21

5. Inhaled air is about 21% oxygen but exhaled air is only about 16% oxygen. By how many percentage points does oxygen fall, and roughly what fraction of the breathed-in oxygen is that? [\[Respiration in Organisms\]](#)
- (A) Falls by 5 points; about three quarters is used
 - (B) Falls by 37 points; all of it is used
 - (C) Falls by 21 points; one quarter is used
 - (D) Falls by 5 points; about one quarter of the oxygen is used
 - (E) Falls by 5 points; about one fifth is used
 - (F) Falls by 16 points; about half is used
6. Two supplementary angles are in the ratio 4 : 5. What is the larger of the two angles? [\[Lines & Angles\]](#)
- (A) 90 degrees
 - (B) 100 degrees
 - (C) 45 degrees
 - (D) 72 degrees
 - (E) 144 degrees
 - (F) 120 degrees
 - (G) 80 degrees
7. Soil X drains 250 mL of water in 5 minutes; soil Y drains 250 mL in 25 minutes. Which soil is more likely sandy, and what are their percolation rates? [\[Soil\]](#)
- (A) X is sandy; X at 5 mL/min and Y at 25 mL/min
 - (B) Y is sandy; X at 10 mL/min and Y at 50 mL/min
 - (C) X is sandy; X drains at 50 mL/min and Y at 10 mL/min
 - (D) X is sandy; X at 1250 mL/min and Y at 6250 mL/min
 - (E) Both drain at the same rate
 - (F) X is clayey; X at 50 mL/min and Y at 10 mL/min
 - (G) Y is sandy; X drains at 50 mL/min and Y at 10 mL/min
8. Simplify the expression $5a + 3b - 2a + 7b$. [\[Algebraic Expressions\]](#)
- (A) $3a + 21b$
 - (B) $10a + 3b$
 - (C) $3a + 10b$
 - (D) $3a + 4b$
 - (E) $13ab$
 - (F) $2a + 10b$
 - (G) $7a + 10b$
9. A cockroach takes in air through tiny holes in its body called spiracles. If it has 10 pairs of spiracles, how many spiracles is that, and which tubes then carry the air inside? [\[Respiration in Organisms\]](#)
- (A) 20 spiracles; air travels through gills
 - (B) 10 spiracles; air travels through lungs
 - (C) 10 spiracles; air travels through tracheae
 - (D) 20 spiracles; air travels through lungs
 - (E) 5 spiracles; air travels through tracheae
 - (F) 20 spiracles; air travels through tracheae

- 10.** Two straight lines cross each other, and one of the four angles formed is 65 degrees. What is the angle vertically opposite it, and an angle right next to it on the line? [\[Lines & Angles\]](#)
- (A) Opposite 25 degrees and next door 155 degrees
 - (B) Opposite 65 degrees and next door 25 degrees
 - (C) Both are 115 degrees
 - (D) 115 degrees opposite and 65 degrees next door
 - (E) Vertically opposite 65 degrees; the next-door angle 115 degrees
 - (F) Both are 90 degrees
 - (G) Both are 65 degrees
- 11.** Arrange the layers of a soil profile in order, starting from the surface and going downwards. [\[Soil\]](#)
- (A) Topsoil, then bedrock, then subsoil, then weathered rock
 - (B) Subsoil, then topsoil, then bedrock, then weathered rock
 - (C) Topsoil, then subsoil, then partly weathered rock, then bedrock
 - (D) Weathered rock, then bedrock, then topsoil, then subsoil
 - (E) Bedrock, then weathered rock, then subsoil, then topsoil
 - (F) Bedrock, then subsoil, then topsoil, then weathered rock
 - (G) Subsoil, then weathered rock, then topsoil, then bedrock
- 12.** In the expression $7x^2y - 3xy + 9$, what is the coefficient of xy , and how many terms does it have? [\[Algebraic Expressions\]](#)
- (A) Coefficient is minus 3; it has two terms
 - (B) Coefficient is 9; it has three terms
 - (C) Coefficient is minus 3; it has three terms
 - (D) Coefficient is 7; it has three terms
 - (E) Coefficient is minus $3xy$; it has three terms
 - (F) Coefficient is 3; it has three terms
- 13.** Water holds far less oxygen than air. If a litre of water holds only about one thirtieth of the oxygen in a litre of air, and air holds about 210 mL of oxygen per litre, how much oxygen is in a litre of water? [\[Respiration in Organisms\]](#)
- (A) About 30 mL per litre
 - (B) About 3 mL per litre
 - (C) About 70 mL per litre
 - (D) About 7 mL per litre
 - (E) About 6300 mL per litre
 - (F) About 35 mL per litre
- 14.** A transversal cuts two parallel lines. One interior angle on one side measures 110 degrees. What is the co-interior (same-side) angle? [\[Lines & Angles\]](#)
- (A) 250 degrees
 - (B) 140 degrees
 - (C) 70 degrees
 - (D) 55 degrees
 - (E) 110 degrees
 - (F) 35 degrees
 - (G) 90 degrees

15. Farmers usually prefer loamy soil for most crops. What makes loamy soil so suitable? [\[Soil\]](#)

- (A) It is made only of the tiniest clay particles
- (B) It contains no humus at all
- (C) It can never hold any water
- (D) It is really just solid rock
- (E) It is pure clay that traps every drop of water
- (F) It is a balanced mix of sand, silt and clay with humus, holding both water and air well

16. Add the two expressions ($2x^2 + 3x - 5$) and ($x^2 - x + 8$). [\[Algebraic Expressions\]](#)

- (A) $3x^2 + 2x + 13$
- (B) $x^2 + 2x + 3$
- (C) $3x^2 - 2x + 3$
- (D) $3x^4 + 2x + 3$
- (E) $3x^2 + 4x + 3$
- (F) $3x^2 + 2x + 3$

17. A newborn baby breathes about 40 times a minute and a calm adult about 16 times a minute. The baby's rate is how many times the adult's, and how many breaths does the baby take in 3 minutes?

[\[Respiration in Organisms\]](#)

- (A) 2.5 times; 120 breaths
- (B) 2 times; 120 breaths
- (C) 2.5 times; 48 breaths
- (D) 3 times; 120 breaths
- (E) 2.4 times; 120 breaths
- (F) 2.5 times; 40 breaths
- (G) 2.5 times; 80 breaths

18. A transversal cuts two parallel lines, making a pair of alternate interior angles equal to $(2x + 10)$ degrees and $(3x - 20)$ degrees. Find x and the size of the angle. [\[Lines & Angles\]](#)

- (A) $x = 10$, angle = 30 degrees
- (B) $x = 30$, angle = 70 degrees
- (C) $x = 30$, angle = 80 degrees
- (D) $x = 2$, angle = 14 degrees
- (E) $x = 6$, angle = 22 degrees
- (F) $x = 50$, angle = 110 degrees

19. One crop needs soil that holds a lot of water, while another needs soil that drains fast. Which soils suit them? [\[Soil\]](#)

- (A) Rice grows in dry sandy soil; cactus-like plants grow in wet clay
- (B) Rice grows well in water-holding clayey soil; plants needing little water suit fast-draining sandy soil
- (C) Both crops grow best in pure gravel
- (D) Rice grows in bedrock; dry-loving plants grow in humus only
- (E) Rice grows well in sandy soil; plants needing little water suit clayey soil
- (F) Both crops need exactly the same soil

20. Subtract $(3a - 4b)$ from $(7a + 2b)$. [\[Algebraic Expressions\]](#)

- (A) $4a - 6b$
- (B) $10a - 2b$
- (C) $4a + 2b$
- (D) $4a - 2b$
- (E) minus $4a + 6b$
- (F) $4a + 6b$

21. Yeast in bread dough respires without oxygen, releasing a gas that makes the dough rise. Name the gas and find the volume released in 45 minutes if the rate is 2 mL each minute. [\[Respiration in Organisms\]](#)

- (A) Oxygen; 90 mL
- (B) Nitrogen; 90 mL
- (C) Carbon dioxide; 90 mL
- (D) Carbon dioxide; 900 mL
- (E) Carbon dioxide; 22.5 mL
- (F) Alcohol; 90 mL

22. Three angles meet at a single point and together sweep all the way around it. Two of them measure 90 degrees and 145 degrees. What is the third angle? [\[Lines & Angles\]](#)

- (A) 55 degrees
- (B) 305 degrees
- (C) 35 degrees
- (D) 145 degrees
- (E) 125 degrees
- (F) 115 degrees
- (G) 235 degrees

23. Arrange the three kinds of soil particle in order from the largest to the smallest. [\[Soil\]](#)

- (A) Sand, then clay, then silt
- (B) Clay, then silt, then sand
- (C) Silt, then clay, then sand
- (D) Sand, then silt, then clay
- (E) Clay, then sand, then humus
- (F) Clay, then sand, then silt
- (G) Silt, then sand, then clay

24. A rectangle has length $(2x + 3)$ and breadth $(x - 1)$. Write its perimeter as a single simplified expression. [\[Algebraic Expressions\]](#)

- (A) $4x + 8$
- (B) $3x + 2$
- (C) $6x + 4$
- (D) $6x + 8$
- (E) $6x - 4$
- (F) $2x$ squared + $x - 3$
- (G) $6x + 2$

25. When oxygen runs short during a hard sprint, muscles break down glucose without it, making a substance that causes cramp. If this only starts after 20 seconds of a 100-second sprint, for how long is it being made, and by what process? [\[Respiration in Organisms\]](#)

- (A) For 80 seconds, by anaerobic respiration
- (B) For 100 seconds, by anaerobic respiration
- (C) For 20 seconds, by anaerobic respiration
- (D) For 5 seconds, by anaerobic respiration
- (E) For 80 seconds, by fermentation in yeast
- (F) For 120 seconds, by anaerobic respiration
- (G) For 80 seconds, by aerobic respiration

26. Two angles form a linear pair. They measure $(x + 30)$ degrees and $2x$ degrees. Find x and the two angles. [\[Lines & Angles\]](#)

- (A) $x = 37.5$; the angles are 67.5 degrees and 75 degrees
- (B) $x = 50$; the angles are 50 degrees and 130 degrees
- (C) $x = 30$; the angles are 60 degrees and 60 degrees
- (D) $x = 50$; the angles are 80 degrees and 100 degrees
- (E) $x = 50$; the angles are 80 degrees and 90 degrees
- (F) $x = 60$; the angles are 90 degrees and 120 degrees
- (G) $x = 25$; the angles are 55 degrees and 50 degrees

27. A soil has a percolation rate of 8 mL per minute. How long will it take for 320 mL of water to drain through it? [\[Soil\]](#)

- (A) 2560 minutes
- (B) 40 minutes
- (C) 4 minutes
- (D) 320 minutes
- (E) 5 minutes
- (F) 312 minutes

28. Find the value of $2xy + x - y$ when $x = 3$ and $y = 5$. [\[Algebraic Expressions\]](#)

- (A) 32
- (B) 26
- (C) 30
- (D) 28
- (E) 13
- (F) 38
- (G) 23

29. At night a plant cannot photosynthesise, so it only respire. If it gives out 12 mL of one gas each hour, how much is given out over an 8-hour night, and which gas is it taking in? [\[Respiration in Organisms\]](#)

- (A) 96 mL of carbon dioxide given out; oxygen taken in
- (B) 1.5 mL of carbon dioxide given out; oxygen taken in
- (C) 96 mL of carbon dioxide given out; carbon dioxide taken in
- (D) 20 mL of carbon dioxide given out; oxygen taken in
- (E) 960 mL of carbon dioxide given out; oxygen taken in
- (F) 96 mL of oxygen given out; carbon dioxide taken in
- (G) 8 mL of carbon dioxide given out; oxygen taken in

30. The supplement of an angle is exactly four times its complement. What is the angle? [\[Lines & Angles\]](#)

- (A) 120 degrees
- (B) 100 degrees
- (C) 75 degrees
- (D) 36 degrees
- (E) 45 degrees
- (F) 60 degrees
- (G) 30 degrees

31. The dark, crumbly material in topsoil that makes it fertile is formed from rotted plant and animal remains. What is it called, and where is most of it found? [\[Soil\]](#)

- (A) Humus; only in the subsoil
- (B) Gravel; mostly in the topsoil
- (C) Humus; mostly in the topsoil
- (D) Sand; mostly in the topsoil
- (E) Humus; mostly in the bedrock
- (F) Silt; mostly in the bedrock
- (G) Clay; mostly in the topsoil

32. How is the expression $4x^2 - 7x + 2$ classified by the number of terms it has? [\[Algebraic Expressions\]](#)

- (A) A four-term expression
- (B) A five-term expression
- (C) Monomial (one term)
- (D) A constant
- (E) Trinomial (three terms)
- (F) A single-term expression
- (G) Binomial (two terms)

33. Different plant parts swap gases through different openings. Which set of pairings is fully correct?

[\[Respiration in Organisms\]](#)

- (A) Leaves use stomata; stems use stomata; roots use lenticels
- (B) Leaves use stomata; stems use gills; roots use lenticels
- (C) Leaves use stomata; stems use lenticels; roots use air gaps in the soil
- (D) Leaves use gills; stems use lenticels; roots use stomata
- (E) Leaves use lenticels; stems use stomata; roots use lungs
- (F) All parts use tiny lungs

34. A transversal makes a pair of corresponding angles. One is 75 degrees and the other is $(x - 15)$ degrees. Since corresponding angles between parallel lines are equal, find x . [\[Lines & Angles\]](#)

- (A) $x = 15$
- (B) $x = 90$
- (C) $x = 105$
- (D) $x = 45$
- (E) $x = 75$
- (F) $x = 60$

- 35.** Soil forms very slowly as rocks are broken into smaller and smaller pieces by wind, water and changing temperature. What is this rock-breaking process called? [\[Soil\]](#)
- (A) Fertilising
 - (B) Condensation
 - (C) Evaporation
 - (D) Photosynthesis
 - (E) Sedimentation
 - (F) Weathering
 - (G) Percolation
- 36.** Find the value of $5p - 2q + 1$ when $p = 2$ and $q = \text{minus } 3$. [\[Algebraic Expressions\]](#)
- (A) 7
 - (B) 5
 - (C) 11
 - (D) minus 1
 - (E) 15
 - (F) 17
- 37.** Each quiet breath moves about 500 mL of air in and out of the lungs. At 15 breaths a minute, what volume of air is moved each minute, and is this 'breathing' or 'cellular respiration'? [\[Respiration in Organisms\]](#)
- (A) 75 litres per minute; this is breathing
 - (B) 0.5 litres per minute; this is breathing
 - (C) 7.5 litres per minute; breathing and respiration are the same thing
 - (D) 7.5 litres per minute; this is cellular respiration
 - (E) 750 mL per minute; this is breathing
 - (F) 7.5 litres per minute; this is breathing
 - (G) 7500 litres per minute; this is breathing
- 38.** Two adjacent angles together make a right angle, and one of them is twice the other. What are the two angles? [\[Lines & Angles\]](#)
- (A) 30 degrees and 90 degrees
 - (B) 36 degrees and 54 degrees
 - (C) 30 degrees and 60 degrees
 - (D) 60 degrees and 120 degrees
 - (E) 45 degrees and 45 degrees
 - (F) 20 degrees and 40 degrees
 - (G) 40 degrees and 50 degrees
- 39.** A bare hillside loses 60 kg of topsoil per square metre each year, but planting trees cuts this loss by 75%. How much topsoil is then lost per square metre each year? [\[Soil\]](#)
- (A) 75 kg
 - (B) 20 kg
 - (C) 15 kg
 - (D) 48 kg
 - (E) 80 kg
 - (F) 0.75 kg

40. Simplify the expression $4xy + 2yz - xy + 3yz$. [\[Algebraic Expressions\]](#)

- (A) $3xy + 5yz$
- (B) $9xyyz$
- (C) $3xy + 5$
- (D) $3xy + yz$
- (E) $6xy + 5yz$
- (F) $3xy + 6yz$

41. When you breathe IN, what happens to the diaphragm and the chest cavity? [\[Respiration in Organisms\]](#)

- (A) The diaphragm domes upward and the chest cavity gets larger
- (B) The diaphragm moves down and flattens, and the chest cavity gets larger
- (C) The diaphragm flattens and the chest cavity gets smaller
- (D) The diaphragm stays still while the cavity gets larger
- (E) The diaphragm moves up and the chest cavity gets smaller
- (F) Nothing happens to the diaphragm at all

42. At exactly 3 o'clock the minute hand and hour hand of a clock form an angle between them. Name the type of angle and give its size. [\[Lines & Angles\]](#)

- (A) Acute angle, 90 degrees
- (B) Right angle, 60 degrees
- (C) Right angle, 90 degrees
- (D) Obtuse angle, 120 degrees
- (E) Straight angle, 180 degrees
- (F) Reflex angle, 270 degrees

43. Of the main soil types, which holds the MOST water and which holds the LEAST? [\[Soil\]](#)

- (A) Gravel holds the most; clay holds the least
- (B) All three hold exactly the same amount
- (C) Loamy soil holds the most; clayey soil holds the least
- (D) Sandy soil holds the most; loamy soil holds the least
- (E) Sandy soil holds the most; clayey soil holds the least
- (F) Clayey soil holds the most; loamy soil holds the least
- (G) Clayey soil holds the most water; sandy soil holds the least

44. Write an algebraic expression for the phrase: 'five less than three times a number x'. [\[Algebraic Expressions\]](#)

- (A) $3 - 5x$
- (B) $5 - 3x$
- (C) $5x - 3$
- (D) $3x - 5$
- (E) x divided by 3, minus 5
- (F) 3 times (x - 5)
- (G) $3x + 5$

45. An earthworm has no lungs and no gills. Through which body part does it take in oxygen, and what must stay true of that part? [\[Respiration in Organisms\]](#)

- (A) Through lungs; they must stay dry
- (B) Through its skin; the skin must stay dry
- (C) Through tracheae; they must stay moist
- (D) Through spiracles; they must stay moist
- (E) Through its skin; the skin must stay warm only
- (F) Through gills; they must stay moist
- (G) Through its skin; the skin must stay moist

46. Two supplementary angles differ from each other by 40 degrees. What is the larger angle? [\[Lines & Angles\]](#)

- (A) 140 degrees
- (B) 80 degrees
- (C) 90 degrees
- (D) 110 degrees
- (E) 130 degrees
- (F) 70 degrees
- (G) 100 degrees

47. A soil is percolation-tested three times, giving rates of 12, 10 and 8 mL per minute. What is the average percolation rate? [\[Soil\]](#)

- (A) 20 mL per minute
- (B) 12 mL per minute
- (C) 30 mL per minute
- (D) 10 mL per minute
- (E) 9 mL per minute
- (F) 11 mL per minute

48. Find the value of $a^2 + b^2$ when $a = 5$ and $b = 12$. [\[Algebraic Expressions\]](#)

- (A) 34
- (B) 119
- (C) 144
- (D) 194
- (E) 289
- (F) 17
- (G) 169

49. Breathing out through limewater turns it milky because exhaled air is rich in one gas. If 5 breaths turn 10 mL of limewater milky, how many breaths are needed for 30 mL, and which gas does it? [\[Respiration in](#)

[Organisms\]](#)

- (A) 15 breaths; oxygen turns it milky
- (B) 6 breaths; carbon dioxide turns it milky
- (C) 45 breaths; carbon dioxide turns it milky
- (D) 15 breaths; carbon dioxide turns it milky
- (E) 10 breaths; carbon dioxide turns it milky
- (F) 150 breaths; carbon dioxide turns it milky
- (G) 15 breaths; nitrogen turns it milky

- 50.** A transversal crosses two parallel lines. The two co-interior angles are $(3x)$ degrees and $(2x)$ degrees. Find x and the larger angle. [\[Lines & Angles\]](#)
- (A) $x = 30$; the larger angle is 90 degrees
 - (B) $x = 20$; the larger angle is 60 degrees
 - (C) $x = 18$; the larger angle is 54 degrees
 - (D) $x = 36$; the larger angle is 108 degrees
 - (E) $x = 36$; the larger angle is 144 degrees
 - (F) $x = 36$; the larger angle is 90 degrees
- 51.** In which layer of the soil do most plant roots grow and most humus collect? [\[Soil\]](#)
- (A) The subsoil layer
 - (B) Nowhere, because roots avoid soil
 - (C) The topsoil (the surface layer)
 - (D) The parent rock layer
 - (E) The bedrock
 - (F) The deepest layer of all
 - (G) The partly weathered rock layer
- 52.** Subtract $(x^2 + 2x - 3)$ from $(4x^2 - x + 5)$. [\[Algebraic Expressions\]](#)
- (A) $3x^2 - 3x + 8$
 - (B) $3x^2 - 3x - 8$
 - (C) $3x^2 - 3x + 2$
 - (D) $3x^2 - x + 8$
 - (E) $3x^2 - 3x + 2$
 - (F) $3x^2 + x + 2$
 - (G) $5x^2 + x + 2$
- 53.** Two cells break down equal amounts of glucose: cell P uses oxygen and cell Q does not. Which cell releases more energy, and which leaves a not-fully-broken-down product? [\[Respiration in Organisms\]](#)
- (A) Cell P releases more energy; cell Q leaves alcohol or lactic acid
 - (B) Cell Q releases more energy; it leaves only carbon dioxide
 - (C) Cell P releases more energy; cell P leaves lactic acid
 - (D) Cell Q releases more energy; cell P leaves alcohol
 - (E) Cell P releases less energy; cell Q leaves nothing
 - (F) Both release equal energy and leave nothing
- 54.** A ray cuts an angle of 124 degrees exactly in half. What is each half-angle, and are the two halves complementary or simply equal? [\[Lines & Angles\]](#)
- (A) 248 degrees each; they are equal
 - (B) 62 degrees each; they are equal, not complementary
 - (C) 62 degrees each; they are complementary
 - (D) 62 degrees each; they are supplementary
 - (E) 56 degrees each; they are equal
 - (F) 124 degrees each; they are equal

55. How does the subsoil usually differ from the topsoil lying above it? [\[Soil\]](#)

- (A) It is simply solid bedrock
- (B) It is where most plant roots grow
- (C) It is exactly the same as the topsoil
- (D) It is just loose, dry sand
- (E) It has less humus, is lighter in colour, and holds more minerals washed down from above
- (F) It contains no minerals at all

56. In the expression $9 - 4x + x^2$, name the constant term and the coefficient of x . [\[Algebraic Expressions\]](#)

- (A) Constant is 9; the coefficient of x is 1
- (B) Constant is 0; the coefficient of x is minus 4
- (C) Constant is minus 4; the coefficient of x is 9
- (D) Constant is 9; the coefficient of x is 4
- (E) Constant is 9; the coefficient of x is minus $4x$
- (F) Constant is 9; the coefficient of x is minus 4

57. At rest a person uses about 250 mL of oxygen each minute. During hard exercise this rate triples. How much oxygen is used in 4 minutes of that exercise? [\[Respiration in Organisms\]](#)

- (A) 1000 mL
- (B) 3 mL
- (C) 750 mL
- (D) 250 mL
- (E) 12000 mL
- (F) 3000 mL
- (G) 2500 mL

58. An angle measures 110 degrees. What is the reflex angle that completes the full turn around its vertex? [\[Lines & Angles\]](#)

- (A) 290 degrees
- (B) 160 degrees
- (C) 200 degrees
- (D) 70 degrees
- (E) 250 degrees
- (F) 180 degrees
- (G) 360 degrees

59. A square field of side 40 m is to be spread with compost at 2 kg for every square metre. How much compost is needed in all? [\[Soil\]](#)

- (A) 1280 kg
- (B) 6400 kg
- (C) 1600 kg
- (D) 3200 kg
- (E) 320 kg
- (F) 160 kg

60. The cost in rupees of n notebooks at 12 rupees each plus one bag costing 15 rupees is $(12n + 15)$.

Find the cost when $n = 7$. [\[Algebraic Expressions\]](#)

- (A) 111 rupees
- (B) 99 rupees
- (C) 105 rupees
- (D) 189 rupees
- (E) 84 rupees
- (F) 27 rupees
- (G) 96 rupees

61. Put the path of breathed-in air in the correct order, from outside to the lungs. [\[Respiration in Organisms\]](#)

- (A) Nasal cavity, then nostrils, then lungs, then windpipe, then bronchi
- (B) Nostrils, then bronchi, then windpipe, then nasal cavity, then lungs
- (C) Windpipe, then nostrils, then bronchi, then nasal cavity, then lungs
- (D) Lungs, then bronchi, then windpipe, then nasal cavity, then nostrils
- (E) Nostrils, then lungs, then windpipe, then bronchi, then nasal cavity
- (F) Nostrils, then nasal cavity, then windpipe, then bronchi, then lungs

62. Two vertically opposite angles measure $(4x - 5)$ degrees and $(3x + 15)$ degrees. Find the value of x .

[\[Lines & Angles\]](#)

- (A) $x = 35$
- (B) $x = 20$
- (C) $x = 10$
- (D) $x = 15$
- (E) $x = 25$
- (F) $x = 5$
- (G) $x = 30$

63. Why does water drain very slowly through clayey soil? [\[Soil\]](#)

- (A) Its particles are spread far apart
- (B) Its particles are huge with big gaps between them
- (C) Its tiny particles are packed tightly with very little space between them
- (D) It is full of large air gaps
- (E) It has no solid particles in it
- (F) It is made of solid rock with no soil
- (G) Clay actively pushes water away

64. Simplify 3 times $(x + 2)$ plus 2 times $(x - 1)$, then find its value when $x = 4$. [\[Algebraic Expressions\]](#)

- (A) $5x + 4$; the value is 44
- (B) $5x + 8$; the value is 28
- (C) $5x + 4$; the value is 20
- (D) $5x$; the value is 20
- (E) $6x + 1$; the value is 25
- (F) $5x + 4$; the value is 24

65. A fish pushes water over its gills 80 times each minute to take in oxygen. How many of these gill movements happen in half an hour? [\[Respiration in Organisms\]](#)

- (A) 1200 movements
- (B) 800 movements
- (C) 160 movements
- (D) 240 movements
- (E) 2400 movements
- (F) 2400000 movements
- (G) 4800 movements

66. Two angles measuring 118 degrees and 62 degrees are placed side by side on a straight line. Do they form a linear pair, and why? [\[Lines & Angles\]](#)

- (A) No, because both angles are obtuse
- (B) No, because together they make 90 degrees
- (C) No, because a linear pair must be 90 degrees each
- (D) Yes, because together they make 360 degrees
- (E) Yes, because together they add up to 180 degrees
- (F) Yes, because the two angles are equal
- (G) It cannot be decided from this information

67. A loamy soil is 40% sand and 40% silt by weight, and the rest is clay. What percentage is clay, and how much clay is in 50 kg of this soil? [\[Soil\]](#)

- (A) 60% clay; 30 kg of clay in 50 kg
- (B) 20% clay; 5 kg of clay in 50 kg
- (C) 20% clay; 1 kg of clay in 50 kg
- (D) 20% clay; 10 kg of clay in 50 kg
- (E) 80% clay; 40 kg of clay in 50 kg
- (F) 10% clay; 5 kg of clay in 50 kg

68. Which of the following can be added to $7ab$ to give a single combined like-term result? [\[Algebraic Expressions\]](#)

- (A) 7
- (B) b a squared
- (C) $7b$
- (D) $7a$
- (E) minus $3ab$
- (F) ab squared

69. After a hard sprint a runner keeps panting heavily for a while. Why does this happen, and what substance built up during the sprint? [\[Respiration in Organisms\]](#)

- (A) To take in oxygen so as to make more lactic acid
- (B) To cool the lungs; no substance builds up
- (C) To get rid of oxygen and remove the lactic acid
- (D) To take in extra oxygen to break down the lactic acid that built up
- (E) To take in extra oxygen to break down the alcohol that built up
- (F) To take in extra carbon dioxide to break down the glucose that built up
- (G) To release nitrogen and break down the water that built up

70. The complement of an angle $(2x)$ degrees is $(x + 30)$ degrees. Find x and the smaller of the two angles.

[Lines & Angles]

- (A) $x = 40$; the smaller angle is 80 degrees
- (B) $x = 20$; the smaller angle is 20 degrees
- (C) $x = 20$; the smaller angle is 50 degrees
- (D) $x = 20$; the smaller angle is 40 degrees
- (E) $x = 15$; the smaller angle is 30 degrees
- (F) $x = 30$; the smaller angle is 60 degrees

71. Which soil-and-crop pairing below is WRONG? [Soil]

- (A) Sandy soil paired with plants needing little water
- (B) Loamy soil paired with vegetables
- (C) Sandy soil paired with paddy (rice)
- (D) Loamy soil paired with wheat
- (E) Clayey soil paired with paddy (rice)
- (F) Loamy soil paired with gram
- (G) Clayey-loam paired with cotton

72. Find the value of $(4x + 8)$ divided by 2 when $x = 5$. [Algebraic Expressions]

- (A) 24
- (B) 28
- (C) 14
- (D) 7
- (E) 18
- (F) 13

73. A calm adult breathes about 16 times a minute. About how many breaths is that in one full day?

[Respiration in Organisms]

- (A) About 384 breaths
- (B) About 230400 breaths
- (C) About 11520 breaths
- (D) About 23040000 breaths
- (E) About 23040 breaths
- (F) About 1440 breaths

74. A transversal cuts two parallel lines. An angle of 70 degrees at the top line corresponds to an angle at the bottom line. What is the angle next to that bottom angle on the same straight line? [Lines & Angles]

- (A) 20 degrees
- (B) 160 degrees
- (C) 35 degrees
- (D) 70 degrees
- (E) 90 degrees
- (F) 110 degrees
- (G) 250 degrees

75. Which of these is NOT a real use or importance of soil? [\[Soil\]](#)

- (A) Soil is home to earthworms and microbes
- (B) Soil anchors plant roots
- (C) Soil stores water for plants
- (D) Soil holds air that roots need
- (E) Soil makes sunlight for plants
- (F) Soil supplies minerals to plants

76. A triangle has sides $(x + 2)$, $(2x - 1)$ and $(x + 4)$. Write its perimeter as a single simplified expression.

[\[Algebraic Expressions\]](#)

- (A) $4x + 5$
- (B) $5x + 5$
- (C) $4x - 5$
- (D) $4x + 6$
- (E) $2x + 5$
- (F) $3x + 5$
- (G) $4x + 7$

77. A fish's gills are split into many thin feathery filaments instead of being one flat sheet. What is the main advantage of this design? [\[Respiration in Organisms\]](#)

- (A) It gives a much larger surface area so oxygen is absorbed faster
- (B) It keeps the gills completely dry
- (C) It helps the fish digest food faster
- (D) It helps the fish see better underwater
- (E) It stores more water inside the fish
- (F) It reduces surface area to save energy
- (G) It makes the fish lighter so it floats

78. A straight angle of 180 degrees is split into three angles in the ratio 2 : 3 : 4. What is the largest of the three angles? [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 40 degrees
- (C) 60 degrees
- (D) 90 degrees
- (E) 72 degrees
- (F) 45 degrees
- (G) 100 degrees

79. Earthworms are often called a farmer's friend. What do they mainly do for the soil? [\[Soil\]](#)

- (A) They make the soil hard and packed
- (B) They turn the soil into solid rock
- (C) They remove all the water from soil
- (D) They eat the farmer's crops
- (E) They kill the roots of plants
- (F) They live only in pure clay
- (G) They burrow through it, mixing and loosening it and adding nutrients

80. Find the value of $10 - 3x$ when $x = 4$. [\[Algebraic Expressions\]](#)

- (A) 22
- (B) 2
- (C) minus 2
- (D) minus 12
- (E) 7
- (F) minus 22

81. In a grasshopper, air tubes carry oxygen straight to the body cells. Compared with humans, what does this mean? [\[Respiration in Organisms\]](#)

- (A) It has gills hidden inside its body
- (B) It cannot do aerobic respiration at all
- (C) The insect's blood does not need to carry oxygen to the cells
- (D) Oxygen reaches its cells only through its skin
- (E) It breathes using lungs like a human
- (F) Its blood carries all the oxygen, just like ours

82. A transversal makes two co-interior angles of 95 degrees and 85 degrees with two given lines. Are the two lines parallel? [\[Lines & Angles\]](#)

- (A) It cannot be decided
- (B) Yes, because both angles are obtuse
- (C) Yes, because the co-interior angles add up to 180 degrees
- (D) No, because the two angles are unequal
- (E) No, because they do not add up to 90 degrees
- (F) No, because parallel lines need equal co-interior angles
- (G) Yes, because they add up to 360 degrees

83. Water of 600 mL drains through a soil sample in half an hour. What is the percolation rate in mL per minute? [\[Soil\]](#)

- (A) 20 mL per minute
- (B) 1200 mL per minute
- (C) 5 mL per minute
- (D) 30 mL per minute
- (E) 600 mL per minute
- (F) 40 mL per minute

84. Which of these sets contains ONLY like terms? [\[Algebraic Expressions\]](#)

- (A) $2x$ squared, minus $5x$ squared, $7x$ squared
- (B) $3a$, $3b$, $3c$
- (C) x squared y , xy squared, xyz
- (D) $2x$, $2x$ squared, $2x$ cubed
- (E) ab , bc , ca
- (F) $2x$ squared, $3xy$, $4y$ squared
- (G) 5, $5x$, $5x$ squared

- 85.** A small insect is sealed in a jar holding 100 mL of air, of which 21 mL is oxygen. Later only 16 mL of oxygen is left. How much oxygen did the insect use, and which gas did it add? [\[Respiration in Organisms\]](#)
- (A) Used 37 mL of oxygen; added carbon dioxide
 - (B) Used 5 mL of oxygen; added oxygen
 - (C) Used 21 mL of oxygen; added carbon dioxide
 - (D) Used 5 mL of oxygen; added nitrogen
 - (E) Used 5 mL of oxygen; added carbon dioxide
 - (F) Used 4 mL of oxygen; added carbon dioxide
- 86.** An angle is exactly one-third of its own supplement. What is the angle? [\[Lines & Angles\]](#)
- (A) 60 degrees
 - (B) 75 degrees
 - (C) 135 degrees
 - (D) 90 degrees
 - (E) 45 degrees
 - (F) 30 degrees
 - (G) 120 degrees
- 87.** Why is the topsoil usually darker in colour than the layers lying below it? [\[Soil\]](#)
- (A) Because it is rich in dark humus from decayed remains
 - (B) Because it is always wetter than below
 - (C) Because it contains much more sand
 - (D) Because plant roots avoid it
 - (E) Because sunlight bleaches the lower layers
 - (F) Because it is made only of clay
- 88.** The number of tiles needed for a hall is given by $(8m + 5n)$. Find the number of tiles when $m = 10$ and $n = 6$. [\[Algebraic Expressions\]](#)
- (A) 160
 - (B) 13
 - (C) 80
 - (D) 130
 - (E) 110
 - (F) 86
- 89.** When you breathe OUT, the ribs move down and in and the diaphragm domes upward. What does this do to the air in the lungs? [\[Respiration in Organisms\]](#)
- (A) Nothing in the lungs moves
 - (B) Oxygen is made inside the lungs
 - (C) Carbon dioxide is pulled in from outside
 - (D) The air just stays still inside
 - (E) The chest space shrinks, pushing air out of the lungs
 - (F) The chest space grows, so air rushes in
 - (G) The lungs expand and pull in fresh air

90. At a point on a straight line, three angles are formed in a row: 50 degrees, $(2y)$ degrees, and 70 degrees. Find y . [\[Lines & Angles\]](#)

- (A) $y = 20$
- (B) $y = 90$
- (C) $y = 30$
- (D) $y = 120$
- (E) $y = 45$
- (F) $y = 15$

91. Why does dumping plastic waste harm the soil? [\[Soil\]](#)

- (A) Plastic increases the percolation rate
- (B) Plastic adds useful humus to the soil
- (C) Plastic makes soil darker and richer
- (D) Plastic does not rot away, so it blocks water and air from reaching the soil
- (E) Plastic is a natural soil mineral
- (F) Plastic feeds the earthworms
- (G) Plastic dissolves into the soil quickly

92. Subtract the sum of $(2a + 3)$ and $(a - 5)$ from $10a$. [\[Algebraic Expressions\]](#)

- (A) $6a + 2$
- (B) $7a + 8$
- (C) $7a - 8$
- (D) $7a + 2$
- (E) $8a + 2$
- (F) $13a - 2$

93. Which word equation correctly shows aerobic respiration in a living cell? [\[Respiration in Organisms\]](#)

- (A) Glucose + oxygen gives carbon dioxide + water + energy
- (B) Glucose + carbon dioxide gives oxygen + water
- (C) Carbon dioxide + water gives glucose + oxygen
- (D) Glucose gives lactic acid + oxygen
- (E) Oxygen + water gives glucose + energy
- (F) Glucose + oxygen gives alcohol + energy
- (G) Glucose gives alcohol + carbon dioxide + energy

94. Which of these angle pairs do NOT form a supplementary pair? [\[Lines & Angles\]](#)

- (A) 95 degrees and 95 degrees
- (B) 100 degrees and 80 degrees
- (C) 45 degrees and 135 degrees
- (D) 120 degrees and 60 degrees
- (E) 90 degrees and 90 degrees
- (F) 150 degrees and 30 degrees
- (G) 10 degrees and 170 degrees

- 95.** A garden bed of area 12 square metres is watered using 9 litres of water for every square metre. How many litres of water are used in all? [\[Soil\]](#)
- (A) 96 litres
 - (B) 21 litres
 - (C) 3 litres
 - (D) 120 litres
 - (E) 1080 litres
 - (F) 1.33 litres
 - (G) 108 litres
- 96.** Find the value of x squared minus y squared when $x = 9$ and $y = 4$. [\[Algebraic Expressions\]](#)
- (A) 5
 - (B) 25
 - (C) 13
 - (D) 169
 - (E) 65
 - (F) 41
- 97.** In a class activity, student A breathes 20 times a minute and student B breathes 25 times a minute. Over 8 minutes, how many MORE breaths does B take than A? [\[Respiration in Organisms\]](#)
- (A) 80 more breaths
 - (B) 200 more breaths
 - (C) 5 more breaths
 - (D) 40 more breaths
 - (E) 160 more breaths
 - (F) 360 more breaths
 - (G) 45 more breaths
- 98.** A transversal cuts two parallel lines so that a pair of corresponding angles are $(5x)$ degrees and $(2x + 30)$ degrees. Find x , then find the complement of that angle. [\[Lines & Angles\]](#)
- (A) $x = 10$; the complement is 130 degrees
 - (B) $x = 6$; the complement is 60 degrees
 - (C) $x = 10$; the complement is 50 degrees
 - (D) $x = 10$; the complement is 40 degrees
 - (E) $x = 30$; the complement is 40 degrees
 - (F) $x = 10$; the complement is 90 degrees
 - (G) $x = 10$; the supplement is 130 degrees
- 99.** Three soils have percolation rates of 25, 12 and 5 mL per minute. Which rate belongs to the most sand-rich soil, and which to the most clay-rich soil? [\[Soil\]](#)
- (A) 12 mL/min is the most sandy; 25 mL/min is the most clayey
 - (B) 25 mL/min is the most clayey; 5 mL/min is the most sandy
 - (C) 25 mL/min is the most sandy; 12 mL/min is the most clayey
 - (D) 5 mL/min is the most sandy; 12 mL/min is the most clayey
 - (E) 5 mL/min is the most sandy; 25 mL/min is the most clayey
 - (F) 25 mL/min is the most sandy; 5 mL/min is the most clayey

100. A taxi charges 40 rupees fixed plus 15 rupees for each kilometre. Write the fare for d kilometres, then find the fare for 12 km and how much of it is the distance charge. [\[Algebraic Expressions\]](#)

- (A) Fare = $40 + 15d$; it is 220 rupees, all of it the distance charge
- (B) Fare = $40 + 15d$; the total is 180 rupees
- (C) Fare = $40 + 15d$; it is 220 rupees, of which 40 rupees is the distance charge
- (D) Fare = $40 + 15d$; it is 220 rupees, of which 180 rupees is the distance charge
- (E) Fare = $55d$; it is 660 rupees
- (F) Fare = $15 + 40d$; it is 495 rupees